KA-PAMANA

A Mobile Application for Interactive Learning and Preservation of Oriental Mindoro's Culture and Heritage

CHAPTERS 1 AND 3

Project documentation for presentation and printing

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Chapter 1 Introduction

This chapter explains the purpose, objectives, scope, and value of KaPamana. The current Android application combines a three-lane runner, Oriental Mindoro destinations and festivals, passport progress, and Hanunó'o script practice. The documentation describes the implemented project and separates working features from work that still needs field validation.

Background of the Project

Oriental Mindoro has place-based stories, festivals, and cultural practices that can be introduced through familiar mobile interactions. KaPamana uses short runner missions and learning activities to give players repeated opportunities to encounter that content. It is an independent educational game, not an official cultural record or a substitute for community-led teaching.

The project began as a migration of BuddyPort assets and game concepts into an Ionic React and Capacitor application. It now includes a dedicated Oriental Mindoro route group, festival adventures, a passport, and a Hanunó'o learning area. The active catalog contains 64 destinations with 407 runner missions; 18 festival destinations contribute 72 of those missions. These counts come from the current catalog and were checked against the live festival records in Supabase on 8 November 2026.

Statement of the Problem

The project addresses a design problem: how can a mobile game introduce Oriental Mindoro places and cultural themes while keeping play, learning, and progress easy to use on a phone? The team framed the work around these questions:

• How can each destination provide short, understandable objectives with visible rewards and progress?

• How can festival artwork and guides add local identity while the player keeps their chosen Buddy as the runner?

• How can script practice let users enter a word, see a mechanical Hanunó'o rendering, trace it, and keep or share the result?

• How can an account restore game progress across devices while the app retains a local save for interrupted connectivity?

General Objective

To develop a playable Android application that uses interactive missions and learning activities to introduce Oriental Mindoro's places and cultural themes.

Specific Objectives

• Provide destination and festival runner missions with a selectable Buddy, objectives, rewards, and a passport record.

• Provide Hanunó'o character lessons, word-entry conversion for script practice, tracing, saved drawings, and image sharing.

• Store game progress locally and synchronize an account-linked save through Supabase when connectivity is available.

• Prepare an installable Android test build and verify core content, save paths, and presentation-ready demonstrations.

Significance of the Project

For learners, the app offers a low-pressure entry point to place names, festivals, and script shapes. For the project team, it is a working example of mobile interface design, game state management, cloud synchronization, and content administration. For educators and reviewers, it gives a demonstrable system whose claims can be checked against code, tests, and live data. Cultural descriptions should be reviewed with knowledgeable local contributors before being presented as authoritative.

Scope and Delimitations

The Android test build covers the active route catalog, three-lane runner controls, Buddy selection, mission results, coins and experience, passport stamps, achievements, learning exercises, and account-linked cloud saves. The Oriental Mindoro group covers Calapan City and 14 municipalities. The festival group has 18 named routes, each with four missions and associated background, item, icon, and guide artwork. Festival guides appear as guides; the chosen Buddy remains playable.

The Hanunó’o converter maps Latin spelling to script symbols using documented rules. This is a spelling exercise, not a translation service or an authenticity check. Saved drawings can be exported as images and shared through the device share sheet; availability of Instagram or Facebook Stories depends on installed apps and their current sharing behavior. Festival imagery is generated artwork informed by festival themes, not documentary photography of an event. The development build has no verified Google Play billing or release signing. Physical-device testing and community review remain necessary before a production release.

Chapter 3 Methodology

This chapter records the implementation approach and the checks that can be reproduced. It does not claim a completed survey, controlled experiment, or user acceptance study. The team built the app iteratively, inspected the resulting features, and used automated checks plus live database queries to confirm specific functions.

Development Approach

The project uses iterative development. Each feature is represented by code and content in the repository, exercised through local tests or build checks, and included in an Android debug build. The mobile shell is Capacitor; the interface is Ionic React and TypeScript. Recovered BuddyPort assets and new Oriental Mindoro content are kept in separate data and asset files so provenance remains visible.

System Architecture

Layer	Implementation	Role
Mobile interface	Ionic React and TypeScript	Screens for adventures, Buddies, passport, learning, settings, and account access
Game logic	TypeScript model and engine	Mission objectives, movement, hazards, rewards, and progression
Local storage	Capacitor Preferences and browser storage	Immediate save and offline continuity
Cloud service	Supabase Auth and PostgreSQL	Account-linked player save and live destination content
Android package	Capacitor Android project	Installable debug APK for mobile testing

A normal play sequence is account sign-in, local save load, catalog selection, mission play, result settlement, local save, then queued cloud upload. When a device links to an existing account, the app checks the cloud row and either restores it or asks which save to keep. The save table uses row-level security so an authenticated player manages their own row.

Feature Implementation

Feature	Current implementation
Runner adventures	64 active destinations and 407 missions in the bundled catalog; three-lane play with objective and result screens.
Oriental Mindoro festivals	18 live festival records, 72 missions, festival backgrounds, keepsakes, icons, and guide artwork.
Buddy and passport	Chosen Buddy runs festival missions; guides identify the festival. Results update rewards, stars, stamps, and best scores.
Hanunó’o learning	18 base characters, word-entry conversion, tracing, saved drawings, image export, and device sharing.
Cloud progress	Account-linked player_saves row with local-first updates, retry after offline changes, and restore on another device.

Data and Security Design

Destination content is stored as structured records containing route identifiers, names, artwork paths, and JSON mission levels. Supabase serves the published catalog to the app. The complete player save is a JSON value in one row per account. Authentication is required for player saves, and row-level policies restrict ordinary players to their own data. The app keeps a local copy so a connection interruption does not immediately erase progress. The account flow and save restoration must be demonstrated with test accounts, not real participant data.

Verification and Testing

The repository includes a TypeScript production build, unit tests, browser end-to-end tests, and an Android build workflow. On 8 November 2026, the live Supabase query returned 18 festival destinations, 72 mission levels, and artwork references for all 18 festivals. The public catalog endpoint returned the same festival and mission counts. The player_saves table and its row-level security were confirmed. The Android workflow completed successfully for commit 3d7152d. These checks show build and data availability; they do not establish physical-device usability or an actual cross-device save restoration.

Demonstration and Evaluation Plan

During the presentation, the team should install the latest debug APK before the session and use a prepared test account. The live sequence should show sign-in, choosing a Buddy, opening an Oriental Mindoro festival, playing a short mission, viewing the result and passport, entering a practice word, tracing and saving it, and opening the device share sheet. A second device or fresh installation can demonstrate restore if it has been rehearsed. The team should record pass or fail for each step and note the phone model, Android version, build commit, and connection used.

Use the included video as a fallback if the live demo fails. Keep one phone with the app installed, a charged power source, the MP4 stored locally, the slide deck stored locally, and a second internet source. Before printing, add any institution-specific format and approval information required by the course.

Project Evidence

The implementation can be inspected in the KaPamana repository: src/App.tsx (app screens and navigation), src/Game.tsx and src/engine.ts (runner), src/LearningStudio.tsx and src/hanunoo.ts (learning), src/cloud/save-sync.ts (cloud progress), src/data/catalog.json (routes and levels), supabase/migrations (database), and .github/workflows/android-apk.yml (Android build). The project repository is https://github.com/cedimscalapan/bp.